All fields in the host-state area correspond to processor registers:

- CR0, CR3, and CR4 (64 bits each; 32 bits on processors that do not support Intel 64 architecture).
- RSP and RIP (64 bits each; 32 bits on processors that do not support Intel 64 architecture).
- Selector fields (16 bits each) for the segment registers CS, SS, DS, ES, FS, GS, and TR. There is no field in the host-state area for the LDTR selector.
- Base-address fields for FS, GS, TR, GDTR, and IDTR (64 bits each; 32 bits on processors that do not support Intel 64 architecture).
- The following MSRs:
  - IA32_SYSENTER_CS (32 bits)
  - IA32_SYSENTER_ESP and IA32_SYSENTER_EIP (64 bits; 32 bits on processors that do not support Intel 64 architecture).
  - IA32_PERF_GLOBAL_CTRL (64 bits). This field is supported only on processors that support the 1-setting of the "load IA32_PERF_GLOBAL_CTRL" VM-exit control.
  - IA32_PAT (64 bits). This field is supported only on processors that support the 1-setting of the "load IA32_PAT" VM-exit control.
  - IA32_EFER (64 bits). This field is supported only on processors that support the 1-setting of the "load IA32_EFER" VM-exit control.
  - IA32_S_CET (64 bits; 32 bits on processors that do not support Intel 64 architecture). This field is supported only on processors that support the 1-setting of the "load CET state" VM-exit control.
  - IA32_INTERRUPT_SSP_TABLE_ADDR (64 bits; 32 bits on processors that do not support Intel 64 architecture). This field is supported only on processors that support the 1-setting of the "load CET state" VM-exit control.
  - IA32_PKRS (64 bits). This field is supported only on processors that support the 1-setting of the "load PKRS" VM-exit control.
- The shadow-stack pointer register SSP (64 bits; 32 bits on processors that do not support Intel 64 architecture). This field is supported only on processors that support the 1-setting of the "load CET state" VM-exit control.

In addition to the state identified here, some processor state components are loaded with fixed values on every VM exit; there are no fields corresponding to these components in the host-state area. See Section 29.5 for details of how state is loaded on VM exits.

26.6 VM-EXECUTION CONTROL FIELDS

The VM-execution control fields govern VMX non-root operation. These are described in Section 26.6.1 through Section 26.6.8.

26.6.1 Pin-Based VM-Execution Controls

The pin-based VM-execution controls constitute a 32-bit vector that governs the handling of asynchronous events (for example: interrupts).[1] Table 26-5 lists the controls. See Chapter 28 for how these controls affect processor behavior in VMX non-root operation.

1. Some asynchronous events cause VM exits regardless of the settings of the pin-based VM-execution controls (see Section 27.2).

Table 26-5. Definitions of Pin-Based VM-Execution Controls

Bit Position(s)	Name	Description
0	External-interrupt exiting	If this control is 1, external interrupts cause VM exits. Otherwise, they are delivered normally through the guest interrupt-descriptor table (IDT). If this control is 1, the value of RFLAGS.IF does not affect interrupt blocking.
3	NMI exiting	If this control is 1, non-maskable interrupts (NMIs) cause VM exits. Otherwise, they are delivered normally using descriptor 2 of the IDT. This control also determines interactions between IRET and blocking by NMI (see Section 27.3).
5	Virtual NMIs	If this control is 1, NMIs are never blocked and the "blocking by NMI" bit (bit 3) in the interruptibility-state field indicates "virtual-NMI blocking" (see Table 26-3). This control also interacts with the "NMI-window exiting" VM-execution control (see Section 26.6.2).
6	Activate VMX-preemption timer	If this control is 1, the VMX-preemption timer counts down in VMX non-root operation; see Section 27.5.1. A VM exit occurs when the timer counts down to zero; see Section 27.2.
7	Process posted interrupts	If this control is 1, the processor treats interrupts with the posted-interrupt notification vector (see Section 26.6.8) specially, updating the virtual-APIC page with posted-interrupt requests (see Section 31.6).

All other bits in this field are reserved, some to 0 and some to 1. Software should consult the VMX capability MSRs IA32_VMX_PINBASED_CTLS and IA32_VMX_TRUE_PINBASED_CTLS (see Appendix A.3.1) to determine how to set reserved bits. Failure to set reserved bits properly causes subsequent VM entries to fail (see Section 28.2.1.1).

The first processors to support the virtual-machine extensions supported only the 1-settings of bits 1, 2, and 4. The VMX capability MSR IA32_VMX_PINBASED_CTLS will always report that these bits must be 1. Logical processors that support the 0-settings of any of these bits will support the VMX capability MSR IA32_VMX_TRUE_PINBASED_CTLS MSR, and software should consult this MSR to discover support for the 0-settings of these bits. Software that is not aware of the functionality of any one of these bits should set that bit to 1.

26.6.2 Processor-Based VM-Execution Controls

The processor-based VM-execution controls constitute three vectors that govern the handling of synchronous events, mainly those caused by the execution of specific instructions.[1] These are the **primary processor-based VM-execution controls** (32 bits), the **secondary processor-based VM-execution controls** (32 bits), and the tertiary **VM-execution controls** (64 bits).

Table 26-6 lists the primary processor-based VM-execution controls. See Chapter 26 for more details of how these controls affect processor behavior in VMX non-root operation.

Table 26-6. Definitions of Primary Processor-Based VM-Execution Controls

Bit Position(s)	Name	Description
2	Interrupt-window exiting	If this control is 1, a VM exit occurs at the beginning of any instruction if RFLAGS.IF = 1 and there are no other blocking of interrupts (see Section 26.4.2).
3	Use TSC offsetting	This control determines whether executions of RDTSC, executions of RDTSCP, and executions of RDMSR that read from the IA32_TIME_STAMP_COUNTER MSR return a value modified by the TSC offset field (see Section 26.6.5 and Section 27.3).
7	HLT exiting	This control determines whether executions of HLT cause VM exits.
9	INVLPG exiting	This determines whether executions of INVLPG and INVPCID cause VM exits.
10	MWAIT exiting	This control determines whether executions of MWAIT cause VM exits.
11	RDPMC exiting	This control determines whether executions of RDPMC cause VM exits.

1. Some instructions cause VM exits regardless of the settings of the processor-based VM-execution controls (see Section 27.1.2), as do task switches (see Section 27.2).

Table 26-6. Definitions of Primary Processor-Based VM-Execution Controls (Contd.)

Bit Position(s)	Name	Description
12	RDTSC exiting	This control determines whether executions of RDTSC and RDTSCP cause VM exits.
15	CR3-load exiting	In conjunction with the CR3-target controls (see Section 26.6.7), this control determines whether executions of MOV to CR3 cause VM exits. See Section 27.1.3. The first processors to support the virtual-machine extensions supported only the 1-setting of this control.
16	CR3-store exiting	This control determines whether executions of MOV from CR3 cause VM exits. The first processors to support the virtual-machine extensions supported only the 1-setting of this control.
17	Activate tertiary controls	This control determines whether the tertiary processor-based VM-execution controls are used. If this control is 0, the logical processor operates as if all the tertiary processor-based VM-execution controls were also 0.
19	CR8-load exiting	This control determines whether executions of MOV to CR8 cause VM exits.
20	CR8-store exiting	This control determines whether executions of MOV from CR8 cause VM exits.
21	Use TPR shadow	Setting this control to 1 enables TPR virtualization and other APIC-virtualization features. See Chapter 31.
22	NMI-window exiting	If this control is 1, a VM exit occurs at the beginning of any instruction if there is no virtual-NMI blocking (see Section 26.4.2).
23	MOV-DR exiting	This control determines whether executions of MOV DR cause VM exits.
24	Unconditional I/O exiting	This control determines whether executions of I/O instructions (IN, INS/INSB/INSW/INSD, OUT, and OUTS/OUTSB/OUTSW/OUTSD) cause VM exits.
25	Use I/O bitmaps	This control determines whether I/O bitmaps are used to restrict executions of I/O instructions (see Section 26.6.4 and Section 27.1.3). For this control, “0” means “do not use I/O bitmaps” and “1” means “use I/O bitmaps.” If the I/O bitmaps are used, the setting of the “unconditional I/O exiting” control is ignored.
27	Monitor trap flag	If this control is 1, the monitor trap flag debugging feature is enabled. See Section 27.5.2.
28	Use MSR bitmaps	This control determines whether MSR bitmaps are used to control execution of the RDMSR and WRMSR instructions (see Section 26.6.9 and Section 27.1.3). For this control, “0” means “do not use MSR bitmaps” and “1” means “use MSR bitmaps.” If the MSR bitmaps are not used, all executions of the RDMSR and WRMSR instructions cause VM exits.
29	MONITOR exiting	This control determines whether executions of MONITOR cause VM exits.
30	PAUSE exiting	This control determines whether executions of PAUSE cause VM exits.
31	Activate secondary controls	This control determines whether the secondary processor-based VM-execution controls are used. If this control is 0, the logical processor operates as if all the secondary processor-based VM-execution controls were also 0.

All other bits in this field are reserved, some to 0 and some to 1. Software should consult the VMX capability MSRs IA32_VMX_PROCBASED_CTLS and IA32_VMX_TRUE_PROCBASED_CTLS (see Appendix A.3.2) to determine how to set reserved bits. Failure to set reserved bits properly causes subsequent VM entries to fail (see Section 28.2.1.1).

The first processors to support the virtual-machine extensions supported only the 1-settings of bits 1, 4–6, 8, 13–16, and 26. The VMX capability MSR IA32_VMX_PROCBASED_CTLS will always report that these bits must be 1. Logical processors that support the 0-settings of any of these bits will support the VMX capability MSR IA32_VMX-_TRUE_PROCBASED_CTLS MSR, and software should consult this MSR to discover support for the 0-settings of these bits. Software that is not aware of the functionality of any one of these bits should set that bit to 1.

Bit 31 of the primary processor-based VM-execution controls determines whether the secondary processor-based VM-execution controls are used. If that bit is 0, VM entry and VMX non-root operation function as if all the secondary processor-based VM-execution controls were 0. Processors that support only the 0-setting of bit 31 of the primary processor-based VM-execution controls do not support the secondary processor-based VM-execution controls.

Table 26-7 lists the secondary processor-based VM-execution controls. See Chapter 26 for more details of how these controls affect processor behavior in VMX non-root operation.

Table 26-7. Definitions of Secondary Processor-Based VM-Execution Controls

Bit Position(s)	Name	Description
0	Virtualize APIC accesses	If this control is 1, the logical processor treats specially accesses to the page with the APIC-access address. See Section 31.4.
1	Enable EPT	If this control is 1, extended page tables (EPT) are enabled. See Section 30.3.
2	Descriptor-table exiting	This control determines whether executions of LGDT, LIDT, LLDT, LTR, SGDT, SIDT, SLDT, and STR cause VM exits.
3	Enable RDTSCP	If this control is 0, any execution of RDTSCP causes an invalid-opcode exception (#UD).
4	Virtualize x2APIC mode	If this control is 1, the logical processor treats specially RDMSR and WRMSR to APIC MSRs (in the range 800H–8FFH). See Section 31.5.
5	Enable VPID	If this control is 1, cached translations of linear addresses are associated with a virtual-processor identifier (VPID). See Section 30.1.
6	WBINVD exiting	This control determines whether executions of WBINVD and WBNOINVD cause VM exits.
7	Unrestricted guest	This control determines whether guest software may run in unpaged protected mode or in real-address mode.
8	APIC-register virtualization	If this control is 1, the logical processor virtualizes certain APIC accesses. See Section 31.4 and Section 31.5.
9	Virtual-interrupt delivery	This controls enables the evaluation and delivery of pending virtual interrupts as well as the emulation of writes to the APIC registers that control interrupt prioritization.
10	PAUSE-loop exiting	This control determines whether a series of executions of PAUSE can cause a VM exit (see Section 26.6.13 and Section 27.1.3).
11	RDRAND exiting	This control determines whether executions of RDRAND cause VM exits.
12	Enable INVPCID	If this control is 0, any execution of INVPCID causes a #UD.
13	Enable VM functions	Setting this control to 1 enables use of the VMFUNC instruction in VMX non-root operation. See Section 27.5.6.
14	VMCS shadowing	If this control is 1, executions of VMREAD and VMWRITE in VMX non-root operation may access a shadow VMCS (instead of causing VM exits). See Section 26.10 and Section 32.3.
15	Enable ENCLS exiting	If this control is 1, executions of ENCLS consult the ENCLS-exiting bitmap to determine whether the instruction causes a VM exit. See Section 26.6.16 and Section 27.1.3.
16	RDSEED exiting	This control determines whether executions of RDSEED cause VM exits.
17	Enable PML	If this control is 1, an access to a guest-physical address that sets an EPT dirty bit first adds an entry to the page-modification log. See Section 30.3.6.
18	EPT-violation #VE	If this control is 1, EPT violations may cause virtualization exceptions (#VE) instead of VM exits. See Section 27.5.7.
19	Conceal VMX from PT	If this control is 1, Intel Processor Trace suppresses from PIPs an indication that the processor was in VMX non-root operation and omits a VMCS packet from any PSB+ produced in VMX non-root operation (see Chapter 35).
20	Enable XSAVES/XRSTORS	If this control is 0, any execution of XSAVES or XRSTORS causes a #UD.
21	PASID translation	If this control is 1, PASID translation is performed for executions of ENQCMD and ENQCMDS. See Section 27.5.8.
22	Mode-based execute control for EPT	If this control is 1, EPT execute permissions are based on whether the linear address being accessed is supervisor mode or user mode. See Chapter 30.

Table 26-7. Definitions of Secondary Processor-Based VM-Execution Controls (Contd.)

Bit Position(s)	Name	Description
23	Sub-page write permissions for EPT	If this control is 1, EPT write permissions may be specified at the granularity of 128 bytes. See Section 30.3.4.
24	Intel PT uses guest physical addresses	If this control is 1, all output addresses used by Intel Processor Trace are treated as guest-physical addresses and translated using EPT. See Section 27.5.4.
25	Use TSC scaling	This control determines whether executions of RDTSC, executions of RDTSCP, and executions of RDMSR that read from the IA32_TIME_STAMP_COUNTER MSR return a value modified by the TSC multiplier field (see Section 26.6.5 and Section 27.3).
26	Enable user wait and pause	If this control is 0, any execution of TPAUSE, UMONITOR, or UMWAIT causes a #UD.
27	Enable PCONFIG	If this control is 0, any execution of PCONFIG causes a #UD.
30	VMM bus-lock detection	This control determines whether assertion of a bus lock causes a VM exit. See Section 27.2.
31	Instruction timeout	If this control is 1, a VM exit occurs if certain operations prevent the processor from reaching an instruction boundary within a specified amount of time. See Section 26.6.24 and Section 27.2.

All other bits in this field are reserved to 0. Software should consult the VMX capability MSR IA32_VMX_PROC-BASED_CTLS2 (see Appendix A.3.3) to determine which bits may be set to 1. Failure to clear reserved bits causes subsequent VM entries to fail (see Section 28.2.1.1).

Bit 17 of the primary processor-based VM-execution controls determines whether the tertiary processor-based VM-execution controls are used. If that bit is 0, VM entry and VMX non-root operation function as if all the tertiary processor-based VM-execution controls were 0. Processors that support only the 0-setting of bit 17 of the primary processor-based VM-execution controls do not support the tertiary processor-based VM-execution controls.

Table 26-8 lists the tertiary processor-based VM-execution controls. See Chapter 26 for more details of how these controls affect processor behavior in VMX non-root operation.

Table 26-8. Definitions of Tertiary Processor-Based VM-Execution Controls

Bit Position(s)	Name	Description
0	LOADIWKEY exiting	This control determines whether executions of LOADIWKEY cause VM exits.
1	Enable HLAT	This control enables hypervisor-managed linear-address translation. See Section 5.5.1.
2	EPT paging-write control	If this control is 1, EPT permissions can be specified to allow writes only for paging-related updates. See Section 30.3.3.2.
3	Guest-paging verification	If this control is 1, EPT permissions can be specified to prevent accesses using linear addresses whose translation has certain properties. See Section 30.3.3.2.
4	IPI virtualization	If this control is 1, virtualization of interprocessor interrupts (IPIs) is enabled. See Section 31.1.6.
5	SEAM guest-physical address width	This control determines the operation of EPT in SEAM non-root operation. See Section 34.3.2.
6	Enable MSR-list instructions	If this control is 0, any execution of RDMSRLIST or WRMSRLIST causes a #UD.
7	Virtualize IA32_SPEC_CTRL	If this control is 1, the operation of the RDMSR and WRMSR instructions is changed when accessing the IA32_SPEC_CTRL MSR. See Section 26.3.

All other bits in this field are reserved to 0. Software should consult the VMX capability MSR IA32_VMX_PROC-BASED_CTLS3 (see Appendix A.3.4) to determine which bits may be set to 1. Failure to clear reserved bits causes subsequent VM entries to fail (see Section 28.2.1.1).